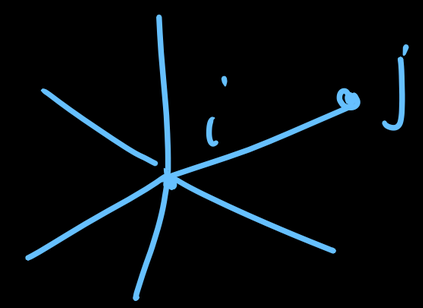
Agenda

→ t-SNE

→ U-MAP

$$p_{ij} = e^{-\frac{\|x_i - x_j\|^2}{2\sigma_i^2}}$$

$$= e^{-d_{ij}^2 / 2\sigma_i^2}$$



$$p_{ij} = \frac{e^{-\frac{\|x_i - x_j\|^2}{2\sigma_i^2}}}{\sum_{k \neq l} e^{-\|x_k - x_l\|^2 / 2\sigma_k^2}}$$

$d \longrightarrow d'$

$x_i \in R^d \qquad y_i \in R^{d'}$

$x_i \qquad x_j$

$x_j \in Neigh(x_i)$

$y_i \qquad y_j$

$y_j \in Neigh(y_i)$

$\underline{\underline{P_{ij}}}$ $\qquad$ $\underline{\underline{Q_{ij}}}$

$\rightarrow$ Preserve pairwise dist in the Neighborhood as best as possible

$$P_{ij} \approx Q_{ij}$$

$$p_{ij} \approx q_{ij}$$

→ Closer in high dim → It should be close in low dim

→ Farther in high dim → Farther in lower dim

p_{ij} = Probab that x_i & x_j are close in high dim

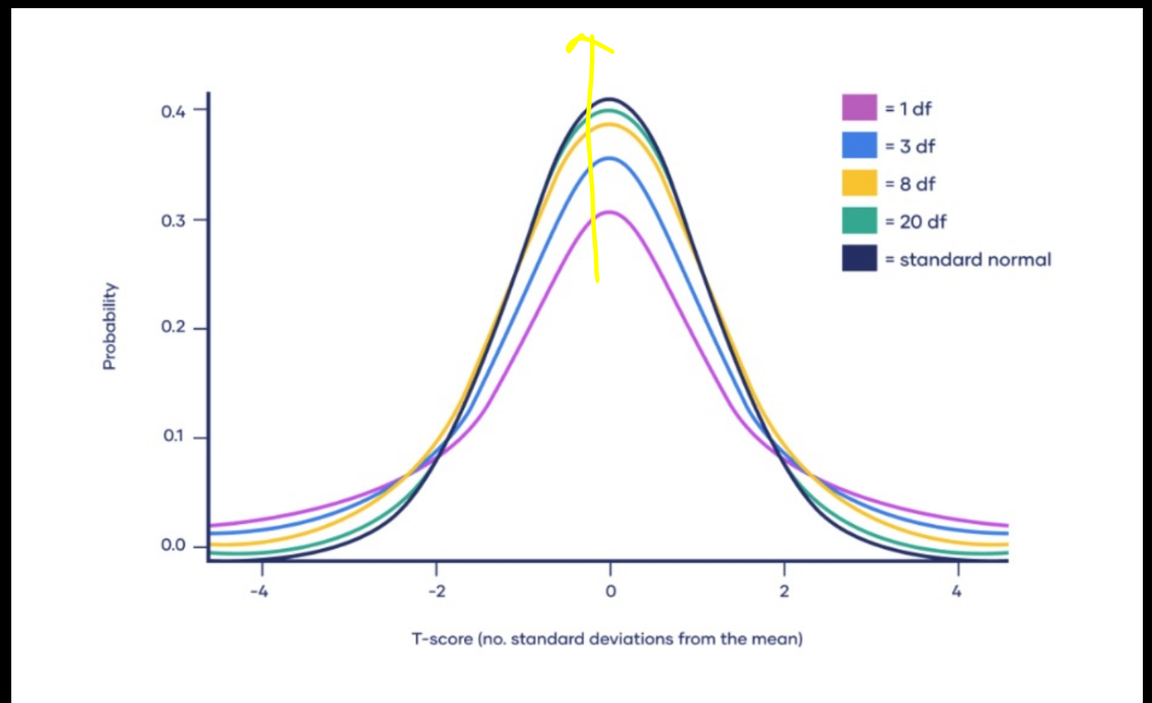
q_{ij} = " y_i & y_j " low dim

t-distribution

→ t-dist is used when population std is unknown

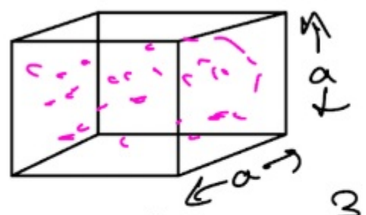
→ t-dist less peak & heavier tails

→ As DOF ↑ t-dist become more peaky & tail decrease



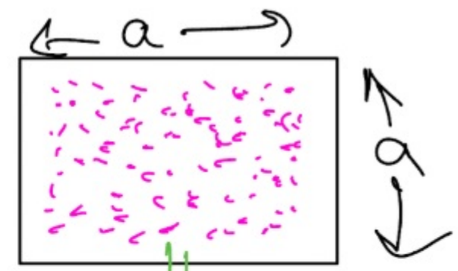
→ Crowding problem

Imagine we have 1 million pts in a cube of side 'a'



$\therefore$ volume $= a^3$

transforming → to 2D (lower dim)



less space hence 1 million datapoints leads to issue called Crowding

High dim

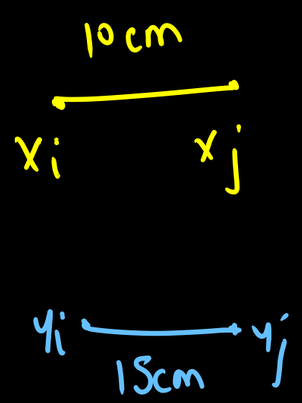
$x_i - x_j$

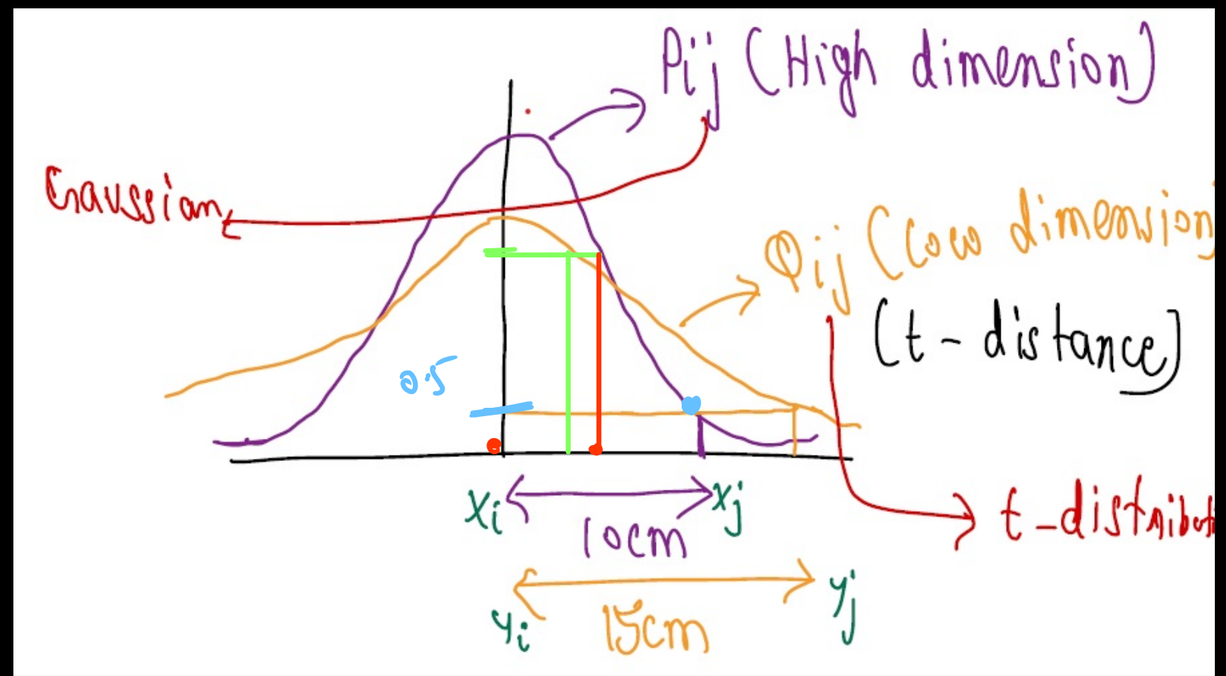
p_{ij}

Low dim

$y_i - y_j$

q_{ij}





t-SNE

$P_{ij} \approx Q_{ij}$

10cm 15cm

$x_i - x_j = 5cm$
$y_i - y_j = 2cm$

Point farther in higher dim will be placed at more distant locations.

t-dist with DoF = 1 (Cauchy distribution)

$$\boxed{p_{ij} \approx q_{ij}}$$

<u>KL-divergence</u>

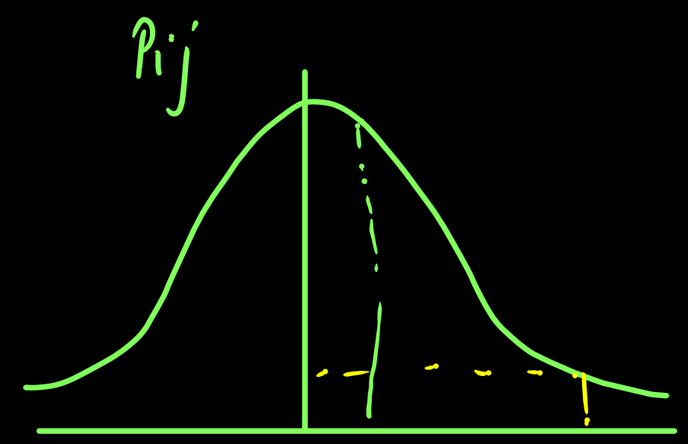
$$KL(P,Q) = \sum\sum p_{ij} \log_e \left(\frac{p_{ij}}{q_{ij}} \right)$$

KL-divergence

$$KL(P,Q) = \sum\sum p_{ij} \log_e \underbrace{\left(\frac{p_{ij}}{q_{ij}}\right)}_{1}$$

$$p_{ij} = q_{ij}$$

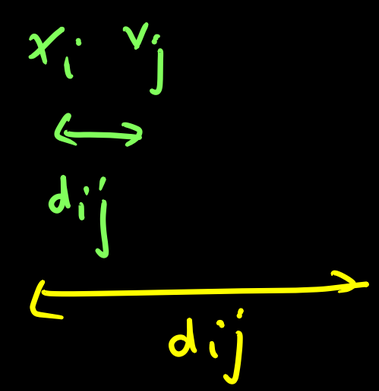
$$p_{ij} = e^{-\frac{\|x_i - x_j\|^2}{2\sigma_i^2}}$$



→ for close point

$d_{ij} \downarrow \quad p_{ij} \uparrow \quad = a_{ij} \uparrow$

$d_{ij} \uparrow \quad p_{ij} \downarrow \quad \approx a_{ij} \downarrow$



Q_{ij} = t-dist with dof = 1

↳ Cauchy dist.

$$\frac{1}{\pi(1+d_{ij}^2)}$$

$$Q_{ij} = \frac{(1+\|y_i - y_j\|^2)^{-1}}{\text{Normalize}}$$

$$\frac{1}{\pi(1+\|y_i - y_j\|^2)}$$

$$= \frac{(1+\|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1+\|y_k - y_l\|^2)^{-1}}$$

Hyperparameter

→ Perplexity → balance out b/w local & global structure

↳ Effective neighbours whose distance we want to preserve

$$e^{-\frac{\|x_i - x_j\|^2}{2\sigma_i^2}}$$

σ_i = Spread

n-neighbours

→ low → Preserve local structure
→ less neighbour.

→ high → large neighbour
→ Preserve global structure.

Limitation of t-SNE

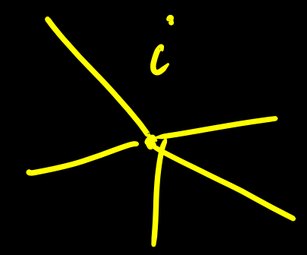
1) Very slow algorithm

2) Very sensitive to perplexity.

$p_{ij} \longrightarrow O(N^2)$

$q_{ij} \longrightarrow$

N



→ Break until 22:19 PM

U-MAP

↳ Came around 2019-2020

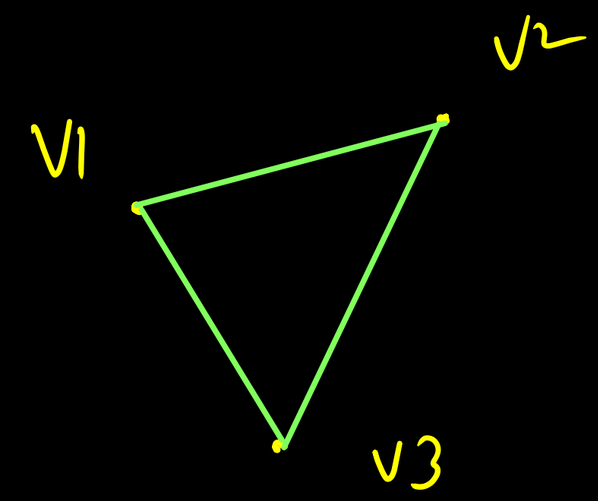
↳ Uniform Manifold Approximation & Projection

Algebric topology

Algorithmic + Optimization

→ Graph data structure

↳ Each point is called vertex/nodes

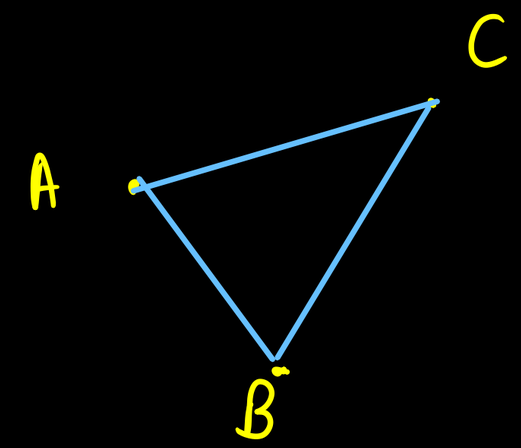


↳ Nodes are connected by edges

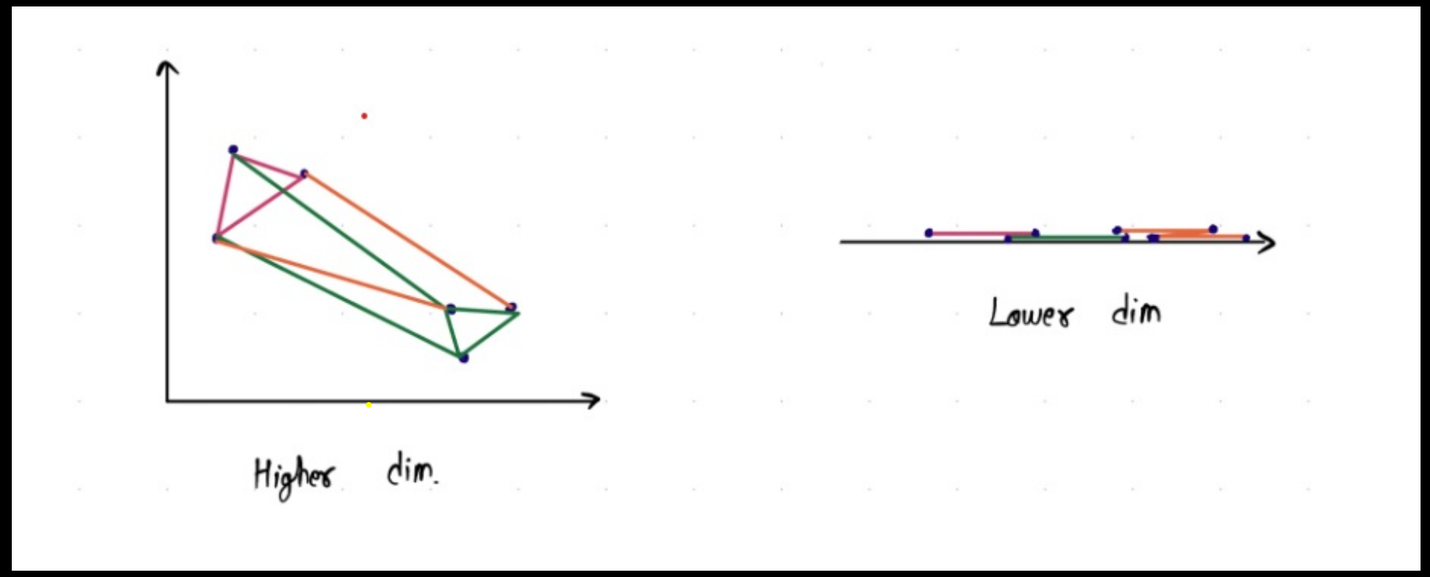
→ Google Map

→ Each of location is a node

→ Street connecting b/w location A & location B is edge

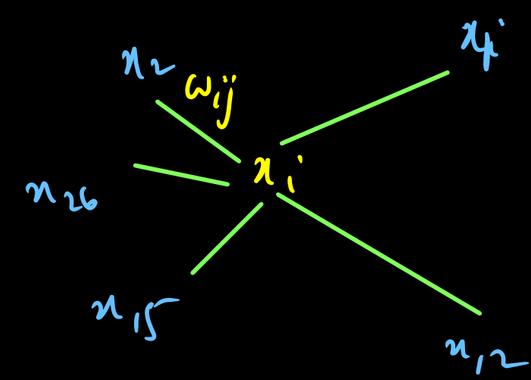


→ We give weights on edge. wT distl


Higher dim.
Lower dim

Algo

1. Find $\underset{\underset{5}{\parallel}}{K}$-NN for x_i



2. Build a graph

$$w_{ij} = \frac{1}{\text{dist}(x_i, x_j)}$$

3. For each $\underset{\text{d-dim}}{x_i} \xrightarrow{\text{Proj}} \underset{\text{2-dim}}{y_i}$

$$\underset{\text{High-dim}}{\underset{\underset{\parallel}{G^{(d)}}}{}} \approx \underset{\text{lower dim}}{\underset{\underset{\parallel}{G^{(2)}}}{}}$$

$\rightarrow$ t-SNE $\quad p_{ij} \approx q_{ij}$

U-MAP $\quad \text{Graph}(x_i) \approx \text{Graph}(y_i)$

Hyperparameter

→ n-neighbor

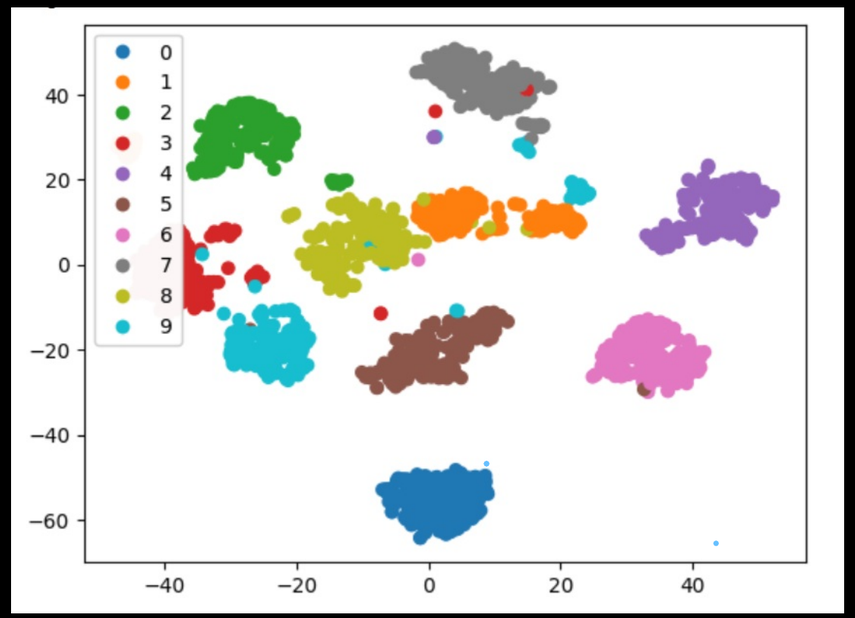
↓ → local structure

↑ → global structure

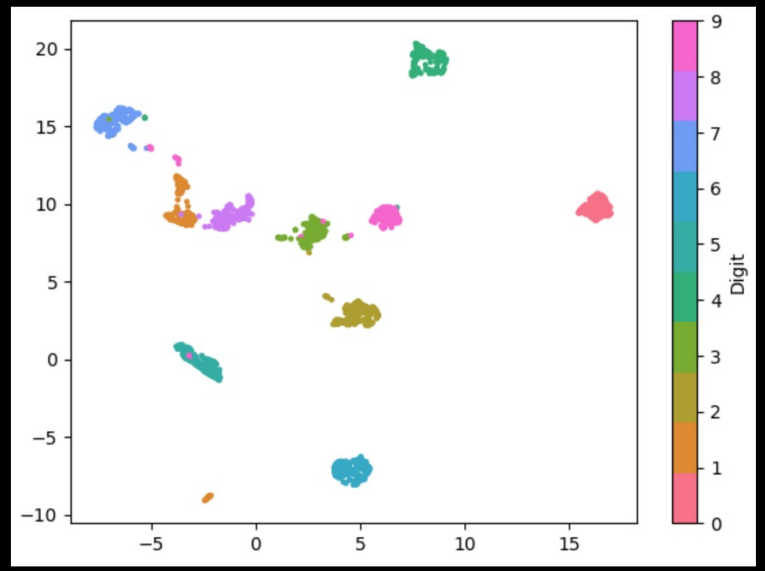
→ min_dist = Min-dist b/w points in low dim

↓ = Points are tightly packed

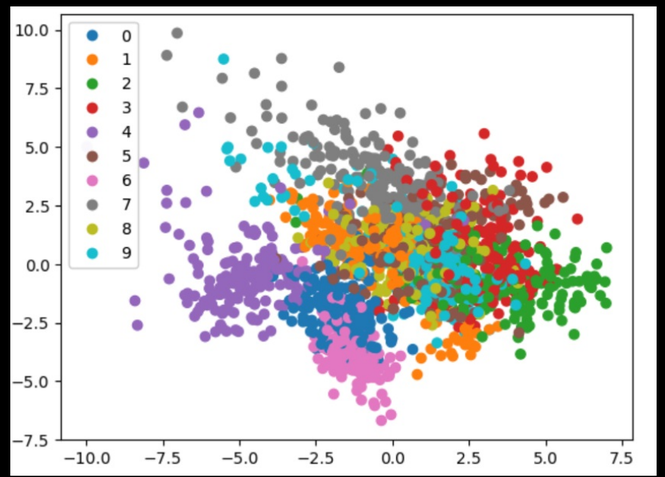
↑ = loosely packed.



t-SNE



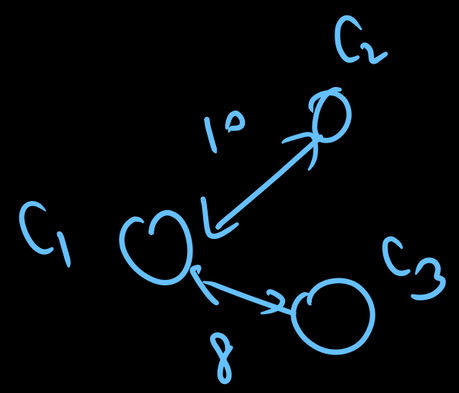
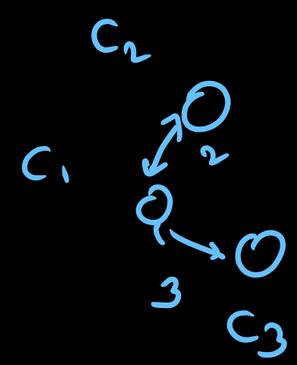
U-MAP



PCA

UMAP > t-SNE > PCA

Summary

1) Hyperparameter

2) Distance b/w clusters are not important.